Marking Calibration — Executive Overview & Rollout Plan

For: Calum Johnson (founder)

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Purpose: what it takes to switch the AI marker from "built but dark" to "live for learners" — across every board and offering — people, scope, sequence and time. Pay is **project-based** (agreed up front per marker), so this plans **scope and sequencing**, not fixed budgets.

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1. The one-paragraph summary

The AI marking engine is built and safe, but it is **deliberately locked per area** until we can prove it agrees with real human examiners *for that board and level*. To unlock each area we need a **calibration set**: real student scripts for that qualification, each independently marked by **two qualified markers**, plus a small set of **annotated exemplars**. The engine marks the same scripts, we measure agreement, and an area only goes live when the AI lands **within half a band/mark of the human markers on at least 80% of scripts**. This is a **per-area data build**, run as **scoped projects** with markers whose fee is **agreed up front**. We roll the areas out in sequence rather than all at once.

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2. Why this is needed (and why it's non-negotiable)

The platform marks children's and fee-paying candidates' work. A confidently-wrong mark damages a learner's exam prep and our reputation. So the engine has a **hard gate in code** (`assertCalibrationGreen`): a live marking route returns "not yet calibrated" (HTTP 503) and shows **nothing** to a learner until a green calibration baseline exists **for that area**. This is not optional polish — it is the trust foundation, and it's already enforced in the request path. The calibration data is the **only** thing standing between "built" and "live", and it is gathered **per board/offering**.

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3. What we are building (per area)

Two assets, both produced by human markers, **for each board/offering we light up**:

ASSET	WHAT IT IS	WHY THE ENGINE NEEDS IT
Calibration set	A bank of scripts for that qualification, each marked **independently by 2 markers** against that board's mark scheme / descriptors	The engine marks the same scripts; we compare AI-vs-human to **prove accuracy** for that area and catch drift. This is the gate.
Exemplar bank	A small set of scripts across the mark/band range, each with a short **expert annotation** explaining *why* it sits there	These are the "anchors" injected into the AI at marking time — the single biggest lever on accuracy. Currently zero exist for any area.

Important: the scripts themselves can be **synthetic** (the platform generates them across the mark range — copyright-clean, no real-student data). The **marks must be human** — we never calibrate the AI against another AI's opinion. Markers grade synthetic scripts exactly as they would real ones.

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4. The gate — the same bar for every area

These are the **real, code-enforced thresholds** (`src/lib/marking/engine/calibration/gate.ts`). An area cannot go live until its calibration set clears all of them:

GATE RULE	MINIMUM
Scripts in the calibration set	≥ **50**
Mark/band levels covered	≥ **2 populated**, spread across the range
Scripts per level	≥ **5**
Marks per script	**2 (independently double-marked)**
Within-half-band/mark agreement	≥ **0.80** (we aim higher)

Recommended commission per area: mark slightly **above** the minimums — around **60 scripts × 2 markers (120 marks)**, full mark/band range, **~10–12 exemplars** — so each baseline is robust, not borderline. Scale up for high-volume flagship areas (e.g. AQA GCSE), trim toward the minimum for lower-volume ones.

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5. The offerings to calibrate (each is its own area)

Every board/level/paper combination is a **separate calibration area** — the AI must be proven for each, because mark schemes differ.

OFFERING	BOARDS / SCOPE	NOTES
GCSE English Language	AQA, Edexcel, OCR, Eduqas	Highest volume; AQA is the priority flagship
GCSE English Literature	AQA, Edexcel, OCR, Eduqas	Set-text essays + unseen poetry/prose
IGCSE English	Cambridge 0500/0990, Edexcel IGCSE (4ET1)	Language + Literature
KS3 English	Criteria-based (no external board)	Extended writing + comprehension
IELTS	Academic + General Training	Writing Task 1 & 2 — the pilot area
EAL	English-as-an-Additional-Language	Writing + language work

You do **not** calibrate all of these at once. Each is a scoped project (or set of projects) with the right qualified markers.

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6. How many markers, and who they need to be

Per area: ~3 qualified markers + 1 senior lead (the lead can be one of the 3 if budget is tight → effectively a 3-person team per area). Markers can cover **multiple areas** if they're qualified for them, so the same people may staff several rollouts.

ROLE	WHAT THEY DO	SKILL SET REQUIRED
Markers (×~3)	The bulk double-marking — each marks ~40 scripts so every script gets 2 of the 3	**Current/recent examiner** for the relevant board (AQA, Edexcel, OCR, Eduqas, Cambridge 0500/0990, Edexcel IGCSE 4ET1), **or** equivalent expert (experienced English teacher marking to that scheme; for IELTS a certified Writing examiner or DELTA/CELTA + IELTS-prep marking; for KS3/EAL an experienced specialist). Native/near-native English, reliable, detail-oriented.
Senior lead	Adjudicates disagreements, writes the exemplar annotations, sanity-checks the set	As above **plus** senior/examiner-trainer judgement to write authoritative per-level annotations.

Why qualified markers specifically: the whole point is to prove the AI matches *examiner* judgement for that board. Marks from a non-examiner aren't a valid reference — they'd give us a green light we couldn't trust. This is the one place we cannot cut the skill bar.

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7. Per-area workload & time

	PER MARKER (×~3)	SENIOR LEAD
Scripts to mark	~40 calibration scripts	~40 (as a marker) + ~15 adjudications + ~10–12 exemplar annotations
Time per script	~12–18 min (marks/bands + brief evidence notes via the form)	same, + ~20 min per exemplar annotation
Total time	**~10–12 hours** each	**~15–18 hours**
Spread	comfortably part-time over **1–2 weeks**	over **~2 weeks**

For a marker, calibrating one area is **less than two normal working days**, done flexibly from home.

8. Cost — project-based, agreed up front

We pay **per project, not a fixed per-script rate.** For each area we scope a defined **quota of scripts and/or services** (e.g. "double-mark a 60-script AQA GCSE Language set + 10 exemplar annotations") and **agree the fee individually, in writing, with each marker before they start.** Once a marker is approved they can mark as much as they like, but are **paid only for the agreed quota and services** per project.

What drives each project fee: the marker's experience, the qualification, and the **volume and complexity** of the scoped work. As a planning anchor, expect **one area ≈ ~135 marks + ~12 annotations** of expert work (Section 7), split across ~3 markers + a lead — sized like a small, well-defined freelance engagement per person.

Ongoing (after an area goes live): a small **top-up** project periodically — a handful of fresh double-marked scripts — keeps that area's calibration honest as the model/prompt evolves. Optional but recommended; it's the early-warning system for drift.

9. End-to-end process (per area)

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1. GENERATE scripts     → platform's synthetic generator produces scripts across the
                          mark/band range for the target board/paper (copyright-clean)
2. RECRUIT + APPROVE    → markers onboarded into /admin/marker-drive; their qualified
                          boards/offerings approved; confidentiality/consent agreed;
                          PROJECT SCOPE + FEE agreed up front per marker
3. CREATE BATCH         → a calibration batch (board + paper) loaded with the scripts
4. ASSIGN (double)      → each script assigned to 2 of the ~3 markers
5. MARK                 → markers grade via their queue: marks/bands per the scheme +
                          brief evidence notes (the structured form)
6. QA + ADJUDICATE      → system computes inter-marker agreement; significant
                          disagreements go to the senior lead; the agreed mark becomes
                          the "human truth"
7. PROMOTE              → QA-passed marks promoted into that area's calibration set;
                          the lead's annotated exemplars promoted into the knowledge
8. CALIBRATE + GO-LIVE  → engine marks the set; agreement is measured; when within-ha
                          ≥0.80 the gate flips green and THAT area opens to learners
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You (or an ops person) drive steps 1–4 and 7–8 from the admin console; the markers do 5–6. **Repeat per area.**

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10. Recommended rollout sequence

Light areas up in order of value and readiness, one (or a few) at a time:

PHASE	AREA(S)	WHY THIS ORDER
1 (pilot)	**IELTS** (Academic + GT, Writing)	Already scoped; proves the whole pipeline end-to-end
2	**GCSE English Language + Literature — AQA**	Highest learner volume; biggest single unlock
3	**GCSE — Edexcel, OCR, Eduqas**	Same qualification, additional boards; markers may overlap
4	**IGCSE** (Cambridge 0500/0990, Edexcel 4ET1)	International reach
5	**KS3 and EAL**	Broaden coverage to the remaining first-class offerings

Each phase is independent: a later area being un-calibrated never blocks an earlier one that's already green. The gate is per-area, so you can ship value incrementally.

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11. What you need to decide / provide

1. **Confirm the rollout sequence** (Section 10) — or re-prioritise by your commercial focus.
2. **Source markers per area** — your network, examiner freelance pools, or marking agencies. (The send-out doc, `marker-brief.md`, recruits across all areas.)

3. **For each project: agree the scope + fee up front** with each marker (quota of scripts/services for an agreed fee).
4. **Approve credentials in the system** — only approved markers, in their approved areas, can mark.

Everything else — the generator, the marker console, the QA maths, the per-area gate — is built and waiting.